TP6- tuning etherchannel

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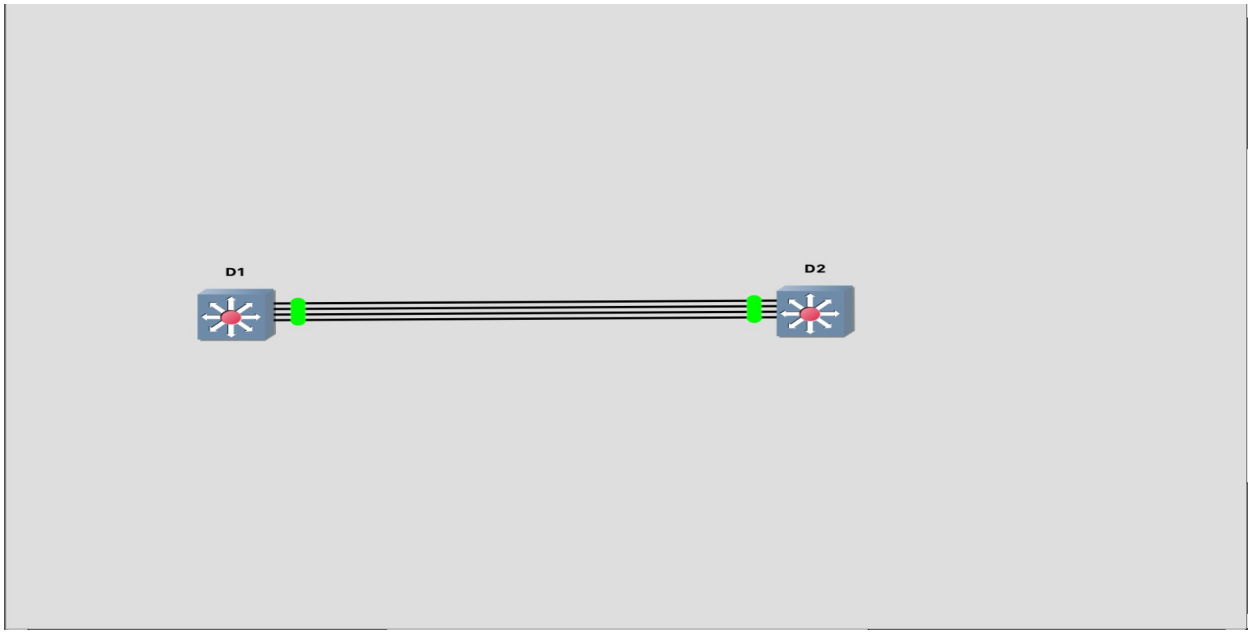
Topology



Bmarco11 — D1 — telnet 192.168.71.128 5000 — 80×24

```
3, changed state to down
D1(config)#interface range e0/0-3
D1(config-if-range)#switchport trunk encapsulation dotq1
                                                       ^
% Invalid input detected at '^' marker.

D1(config-if-range)#switchport trunk encapsulation dot1q
D1(config-if-range)#switchport mode trunk
D1(config-if-range)#no shutdown
D1(config-if-range)#
*Oct 27 10:03:55.993: %LINK-3-UPDOWN: Interface Ethernet0/0, changed state to up
*Oct 27 10:03:55.993: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to up
*Oct 27 10:03:55.993: %LINK-3-UPDOWN: Interface Ethernet0/2, changed state to up
*Oct 27 10:03:55.998: %LINK-3-UPDOWN: Interface Ethernet0/3, changed state to up
*Oct 27 10:03:56.999: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
0, changed state to up
*Oct 27 10:03:56.999: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
1, changed state to up
*Oct 27 10:03:56.999: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to up
*Oct 27 10:03:56.999: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to up
D1(config-if-range)#exit
D1(config)#
```

Bmarco11 — D2 — telnet 192.168.71.128 5001 — 80×24

```
*Oct 27 10:00:49.924: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/
2, changed state to down
*Oct 27 10:00:49.924: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet3/
3, changed state to down
D2(config)#interface range e0/0-3
D2(config-if-range)#switchport trunk encapsulation dot1q
D2(config-if-range)#switchport mode trunk
D2(config-if-range)#no shutdown
D2(config-if-range)#
*Oct 27 10:03:52.743: %LINK-3-UPDOWN: Interface Ethernet0/0, changed state to up
*Oct 27 10:03:52.743: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to up
*Oct 27 10:03:52.744: %LINK-3-UPDOWN: Interface Ethernet0/2, changed state to up
*Oct 27 10:03:52.744: %LINK-3-UPDOWN: Interface Ethernet0/3, changed state to up
D2(config-if-range)#
*Oct 27 10:03:54.758: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
0, changed state to up
*Oct 27 10:03:54.758: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
1, changed state to up
*Oct 27 10:03:54.758: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to up
*Oct 27 10:03:54.758: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to up
D2(config-if-range)#exit
D2(config)#
```

```
Bmarco11 — D2 — telnet 192.168.71.128 5001 — 80×24
D2(config-if-range)#
*Oct 27 10:03:54.758: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
0, changed state to up
*Oct 27 10:03:54.758: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
1, changed state to up
*Oct 27 10:03:54.758: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to up
*Oct 27 10:03:54.758: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to up
D2(config-if-range)#exit
D2(config)#end
D2#
*Oct 27 10:04:44.229: %SYS-5-CONFIG_I: Configured from console by console
D2#show lacp sys-id
32768, aabb.cc80.0200
D2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
D2(config)#lacp system-priority 1
D2(config)#exit
D2#
*Oct 27 10:08:33.576: %SYS-5-CONFIG_I: Configured from console by console
D2#show lacp sys-id
1, aabb.cc80.0200
D2#
```

```
Bmarco11 — D1 — telnet 192.168.71.128 5000 — 80×24
  port-channel        Port Channel interface subcommands
  priority-group      Assign a priority group to an interface
  random-detect       Enable Weighted Random Early Detection (WRED) on an
                      Interface
  rate-limit          Rate Limit
  routing             Per-interface routing configuration
  sap-priority        Assign a priority group
  service             Configure Ether Service
  service-policy      Configure CPL Service Policy
  shutdown            Shutdown the selected interface
  snapshot            Configure snapshot support on the interface
  snmp                Modify SNMP interface parameters
  source              Get config from another source
  spanning-tree       Spanning Tree Subsystem
  subscriber          Subscriber inactivity timeout value.
  switchport          Set switching mode characteristics
  timeout             Define timeout values for this interface
  topology            Configure routing topology on the interface
  traffic-shape       Enable Traffic Shaping on an Interface or Sub-Interface
  transmit-interface  Assign a transmit interface to a receive-only interface
  tx-ring-limit       Configure PA level transmit ring limit
  vtp                 Enable VTP on this interface

D1(config-if)#
```

<u>NOTE: min-links not supported by the cisco image of the switch provided,</u> see config on the screenshot above.

Bmarco11 — D2 — telnet 192.168.71.128 5001 — 80×24

```
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      N - not in use, no aggregation
        f - failed to allocate aggregator

        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
12     Po12(SD)        LACP      Et0/0(s)    Et0/1(s)    Et0/2(s)
                                 Et0/3(s)
[                                                                              ]
D2#
```

Bmarco11 — D2 — telnet 192.168.71.128 5001 — 80×24

```
        A - formed by Auto LAG


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
12     Po12(SD)        LACP      Et0/0(s)    Et0/1(s)    Et0/2(s)
                                 Et0/3(s)
[                                                                              ]
[D2#show lacp internal                                                         ]
Flags:  S - Device is requesting Slow LACPDUs
        F - Device is requesting Fast LACPDUs
        A - Device is in Active mode       P - Device is in Passive mode

Channel group 12
                            LACP port     Admin     Oper    Port        Port
Port      Flags   State     Priority      Key       Key     Number      State
Et0/0     SA      susp      32768         0xC       0xC     0x1         0x7D
Et0/1     SA      susp      32768         0xC       0xC     0x2         0x7D
Et0/2     SA      susp      32768         0xC       0xC     0x3         0x7D
Et0/3     SA      susp      32768         0xC       0xC     0x4         0x7D
D2#
```

Bmarco11 — D2 — telnet 192.168.71.128 5001 — 80×24

```
                                Et0/3(s)
D2#show lacp internal
Flags:  S - Device is requesting Slow LACPDUs
        F - Device is requesting Fast LACPDUs
        A - Device is in Active mode       P - Device is in Passive mode

Channel group 12
                            LACP port     Admin     Oper    Port        Port
Port      Flags   State     Priority      Key       Key     Number      State
Et0/0     SA      susp      32768         0xC       0xC     0x1         0x7D
Et0/1     SA      susp      32768         0xC       0xC     0x2         0x7D
Et0/2     SA      susp      32768         0xC       0xC     0x3         0x7D
Et0/3     SA      susp      32768         0xC       0xC     0x4         0x7D
D2#show etherchannel load-balance
EtherChannel Load-Balancing Configuration:
        src-dst-ip

EtherChannel Load-Balancing Addresses Used Per-Protocol:
Non-IP: Source XOR Destination MAC address
  IPv4: Source XOR Destination IP address
  IPv6: Source XOR Destination IP address

D2#
```